INCH-POUND

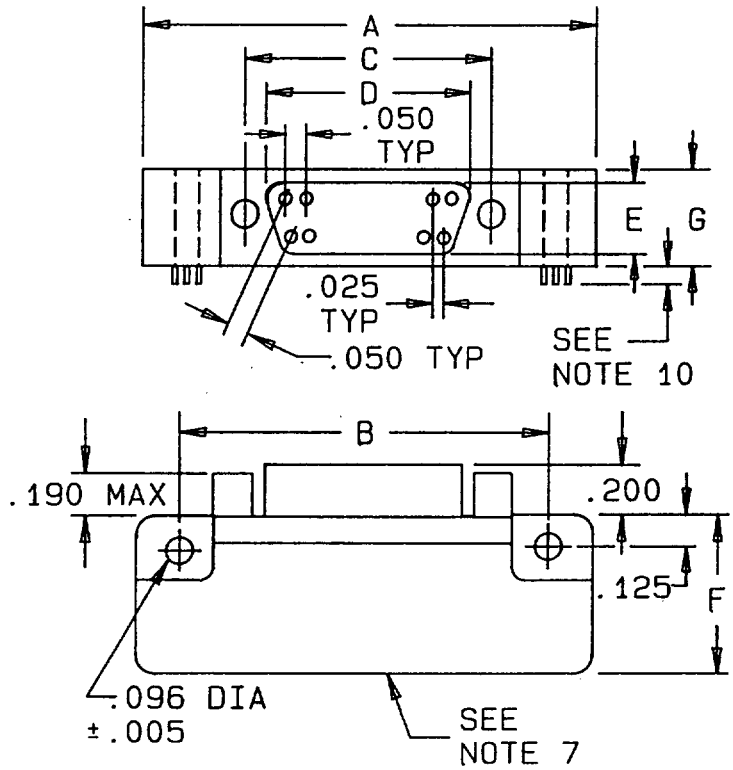
MIL-C-83513/19
27 Feb 1992

MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, RECEPTACLE, MICROMINIATURE, POLARIZED SHELL, RIGHT ANGLE, SOCKET CONTACTS, 2 ROW ,SOLDER TYPE, STANDARD PROFILE, 9 THROUGH 37 CONTACTS, PRINTED CIRCUIT BOARD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification Listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83513.



Inch	mm
.005	0.13
.015	0.38
.025	0.64
.050	1.27
.05	1.3
.096	2.44
.109	2.77
.125	3.18
.150	3.81
.190	4.83
.200	5.08

FIGURE 1. Connector, receptacle, .050 spacing.

AMSC N/A FSC 5935

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:29Z
Check the source to verify that this is the current version before use.

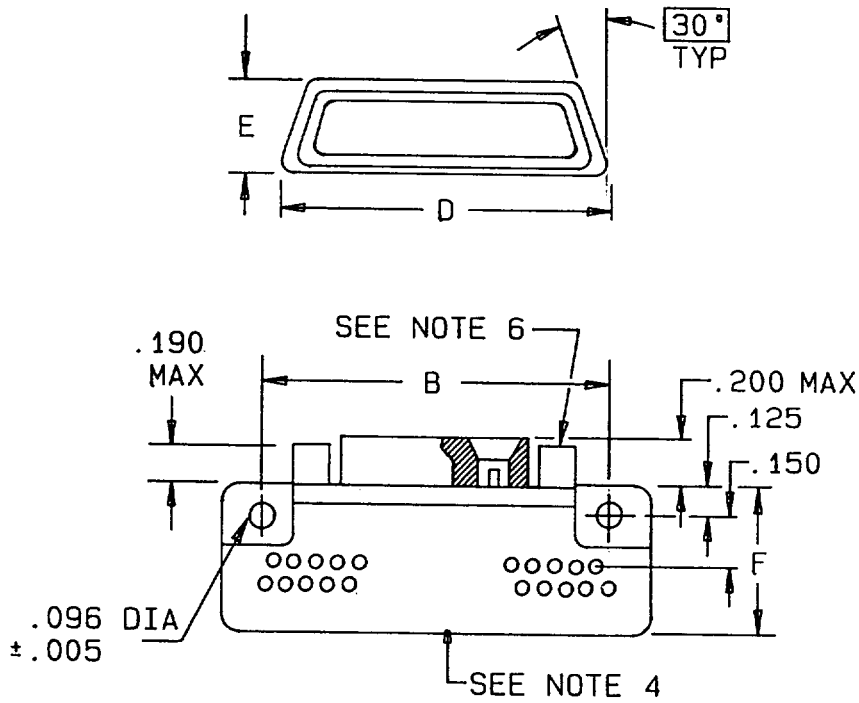


FIGURE 1. Connector, receptacle, .050 spacing - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:29Z
Check the source to verify that this is the current version before use.

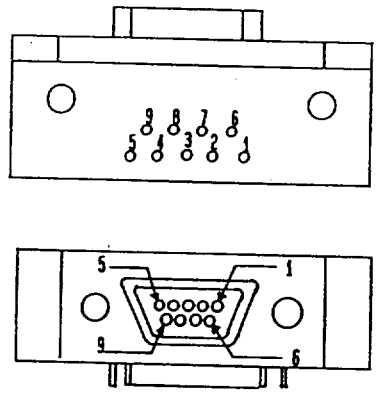
Number of contacts	A Max	B ±.007 0.18 mm	C ±.005 0.13 mm	D Min	E Max	F Max	G Max
9	1.390 (35.31)	1.150 (29.21)	.565 (14.35)	.402 (10.21)	.253 (6.43)	.455 (11.56)	.303 (7.70)
15	1.540 (39.12)	1.30 (33.02)	.715 (18.16)	.552 (14.02)	.253 (6.43)	.455 (11.56)	.303 (7.70)
21	1.690 (40.64)	1.450 (36.83)	.865 (21.97)	.702 (17.83)	.253 (6.43)	.455 (11.56)	.303 (7.70)
25	1.790 (45.47)	1.550 (39.37)	.965 (24.51)	.802 (20.37)	.253 (6.43)	.455 (11.58)	.303 (7.70)
31	2.040 (51.82)	1.80 (45.72)	1.115 (28.32)	.952 (24.18)	.253 (6.43)	.455 (11.56)	.303 (7.70)
37	2.340 (59.44)	2.10 (53.34)	1.265 (32.13)	1.102 (27.99)	.253 (6.43)	.455 (11.56)	.303 (7.70)

NOTES:

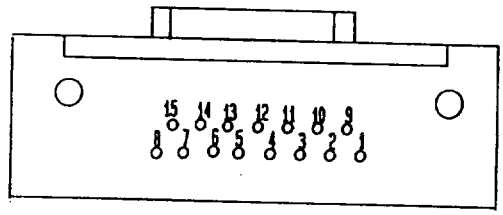
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm).
4. Termination organization cavity to be optionally molded or filled with a potting fill material capable of passing the electrical and environmental requirements of MIL-C-83513.
5. Metal shell shall be of material in accordance with MIL-C-83513 for for class M parts.
6. Jackpost (permanently attached): Corrosion resistant steel in accordance with ASTM A484 and ASTM A582, 300 series stainless steel, passivated in accordance with QQ-P-35.
7. Plastic body shall conform to the requirements of type GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519.
8. Wire termination sockets shall conform to QQ-W-343, no. 24 AWG copper.
9. Interfacial seals are included with each type receptacle connector.
10. Termination lengths available: .109 (2.77 mm), .140 (3.56 mm), or .172 (4.37 mm). The tolerance shall be ±.015 (0.381 mm) for all termination lengths.

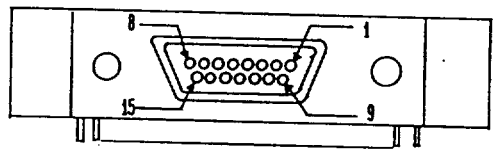
FIGURE 1. Connector, receptacle, .050 spacing. - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:29Z
Check the source to verify that this is the current version before use.

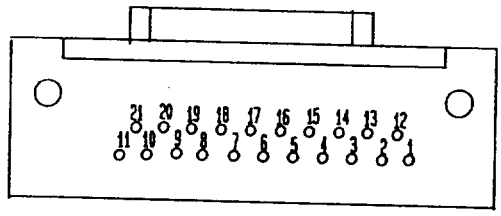


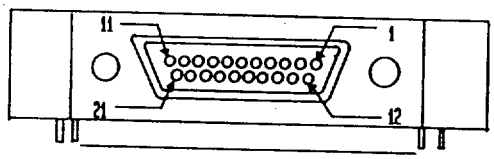
9 CAVITY





15 CAVITY





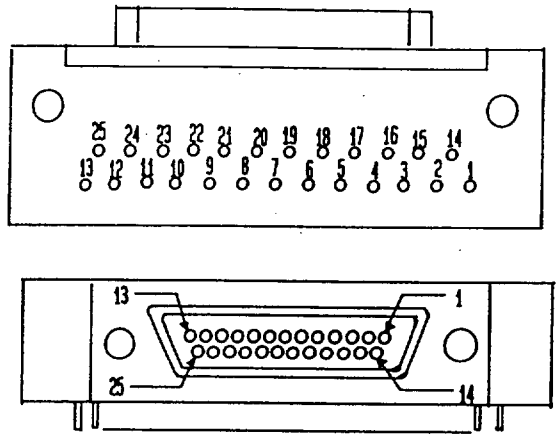
21 CAVITY

NOTE:

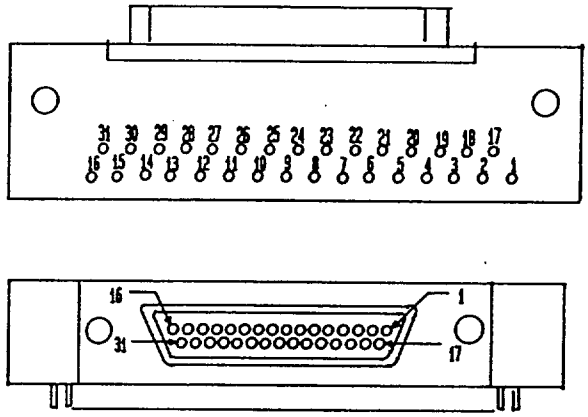
Engaging face of socket insert shown, cavity identification numbers are for reference only and do not appear on the part.

FIGURE 2. Insert arrangement.

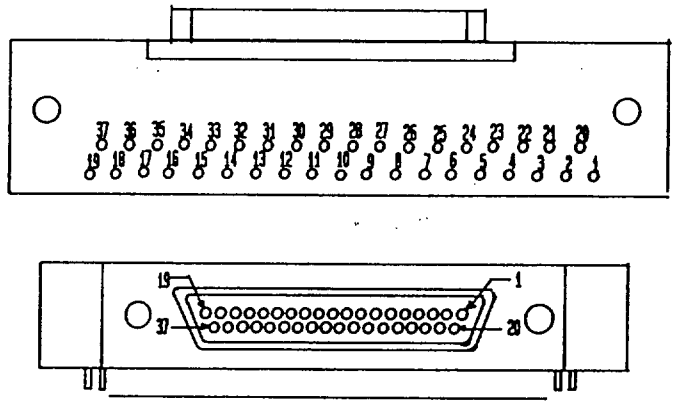
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:29Z
Check the source to verify that this is the current version before use.



25 CAVITY



31 CAVITY



37 CAVITY

NOTE:

Engaging face of socket insert shown, cavity identification numbers are for reference only and do not appear on the part.

FIGURE 2. Insert arrangement - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:29Z
Check the source to verify that this is the current version before use.

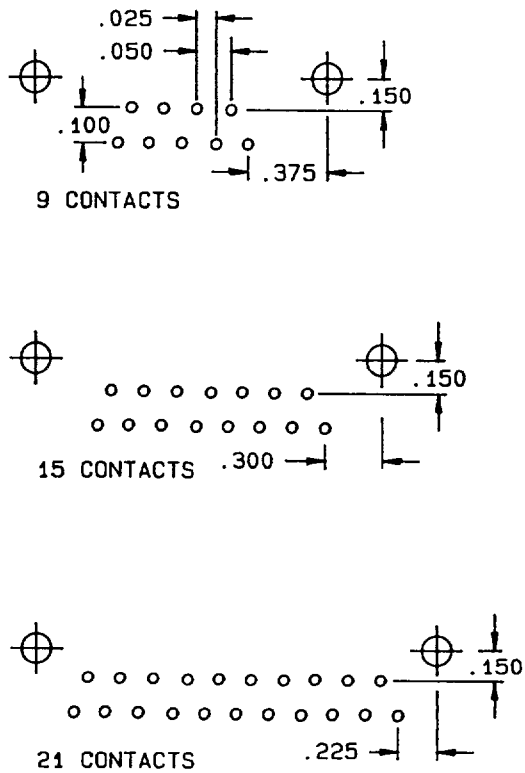
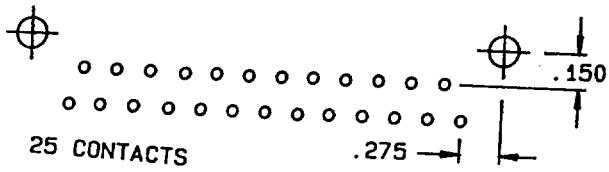
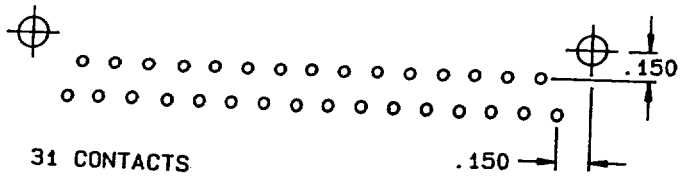


FIGURE 3. Layout arrangement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:29Z
Check the source to verify that this is the current version before use.





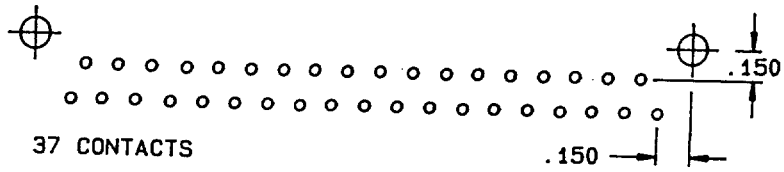


FIGURE 3. Layout arrangement - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:29Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configurations: See figures 1 and 2.

Current rating, maximum: 3 amperes per contact.

Materials: The requirments for shell materials shall be in accordance with MIL-C-83513.

Mating plug: Shall conform to MIL-C-83513/1 and MIL-C-83513/3.

Plating of termination leads: Solder dipping socket of termination leads will be accomplished in SN60 or SN63 in accordance with QQ-S-571.

Part identification number: The PIN shall consists of the letter M, the basic number of the specification sheet, a letter from the insert, a numerical code for the termination length, and a letter code for the shell finish and hardware column.

M83513/19-	D	01	C	P
Specification sheet number	Insert arrangements (see figure 2) A=09 B=15 C=21 D=25 E=31 F=37	Termination length 01=.109 02=.140 03-.172	Shell finish C= cadmium N=electroless nickel (space application)	Hardware N=no jackpost P=jackpost attached T=threaded insert

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, SH
Air Force - 11, 17, 99

User activites:
Army - AT, ME
Navy - CG, MC

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project: 5935-3739-10)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:29Z
Check the source to verify that this is the current version before use.